
MODEL SELECTION AND VALIDATION REPORT (other_fd_biomass model)

OBSERVATIONS:3329

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (other_fd_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.04	0.21
ch_fric	0.12	0.02
ch_feve	-0.15	0.01
teve	-0.44	0.02
sbt_mean	0.33	0.05
sic_mean	0.08	0.05
sbt_sd	0.12	0.03
sst_sd	0.04	0.03
log_chla_mean	0.02	0.02
depth	0.02	0.03

Dispersion parameter: 0.72

Spatiotemporal AR1 correlation (rho): 0.91

Matérn range: 330.42

Spatiotemporal marginal AR1 SD: 0.79

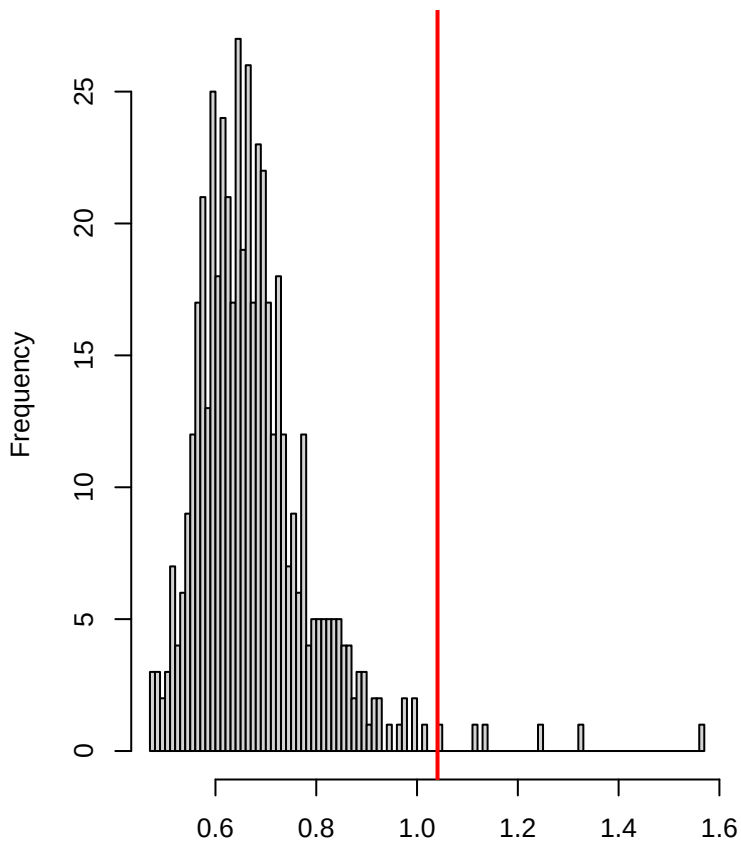
REML criterion at convergence: 26471.363

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

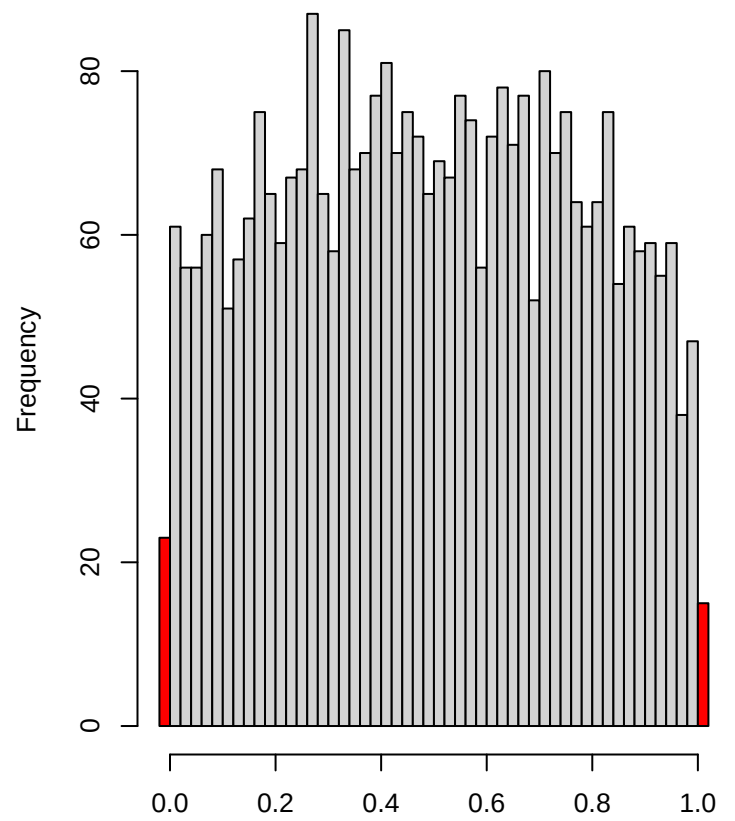
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	14	52970.73	0.3023435
spatiotemporal_random_fields_ar1_with_anisotropy	16	52971.03	246.9179568
spatiotemporal_random_fields_iid_with_anisotropy	15	53217.95	3.2257863
spatiotemporal_random_fields_iid	13	53221.17	147.0437613
spatial_random_fields	13	53368.22	2.1209535
spatial_random_fields_with_anisotropy	15	53370.34	528.3522631
error_only	11	53898.69	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



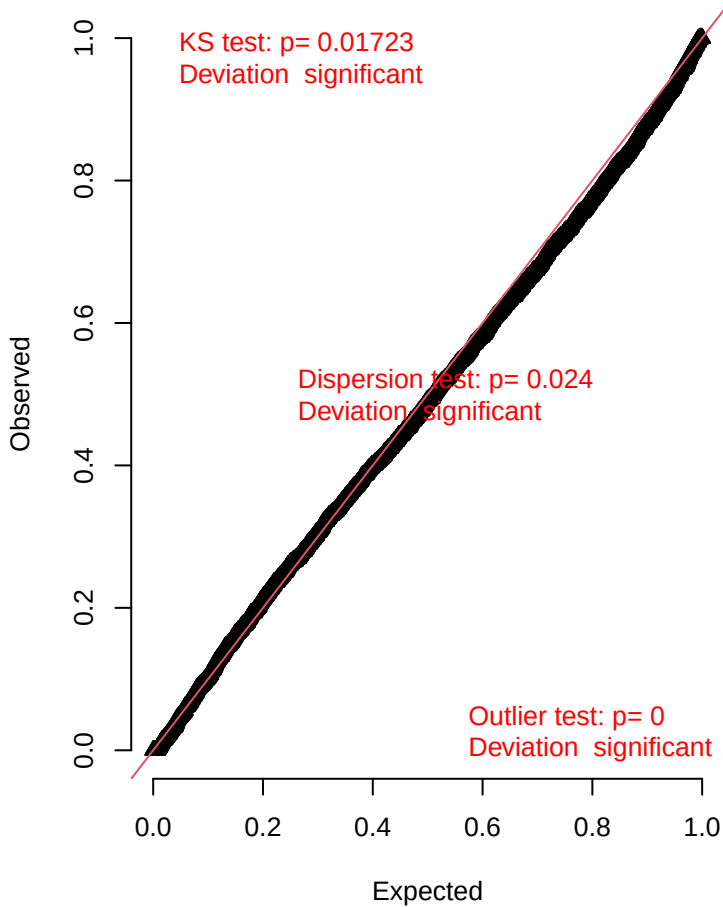
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

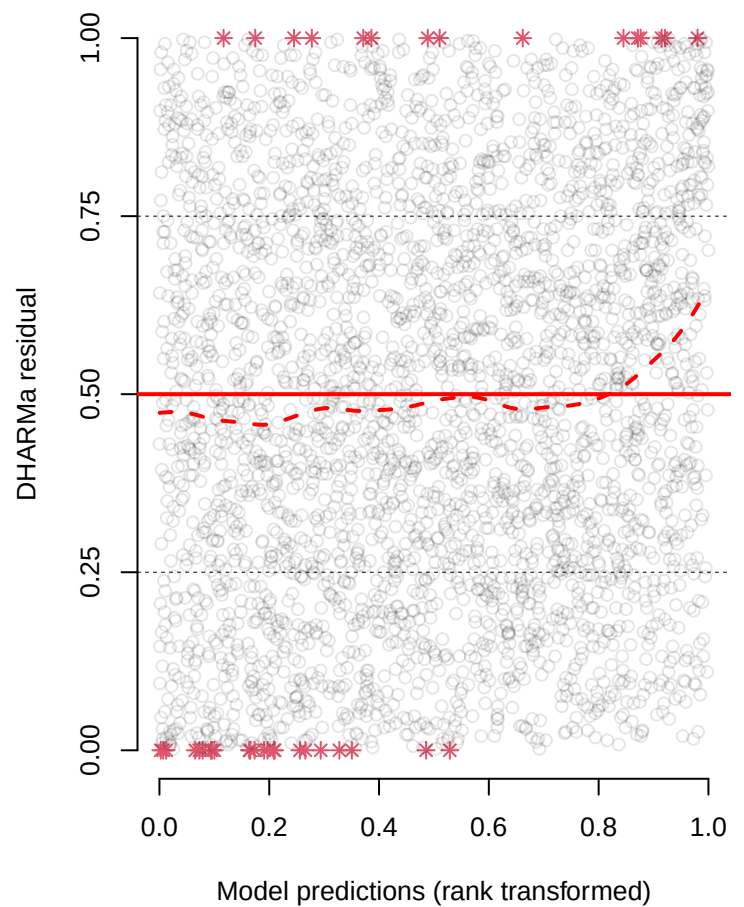


Residuals (outliers are marked red)

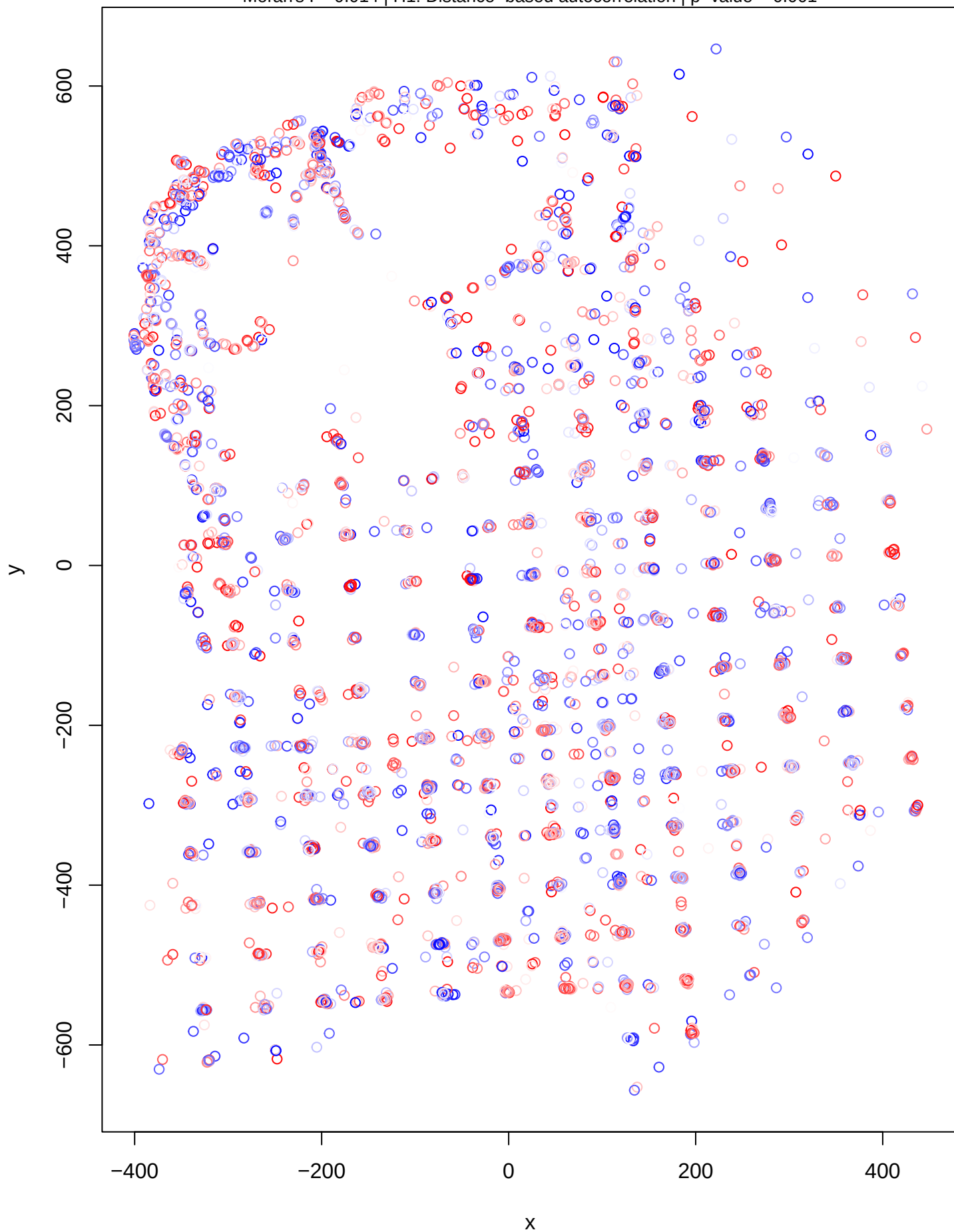
QQ plot residuals



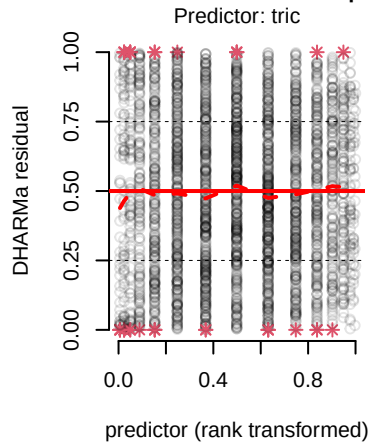
DHARMa residual vs. predicted



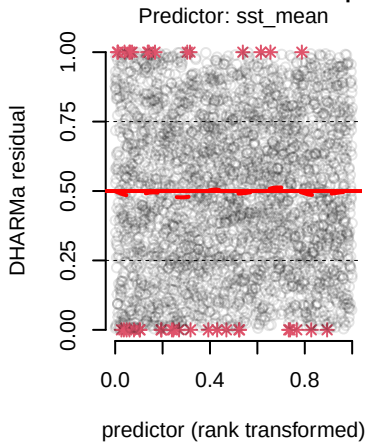
DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.014 | H1: Distance-based autocorrelation | p-value = 0.001



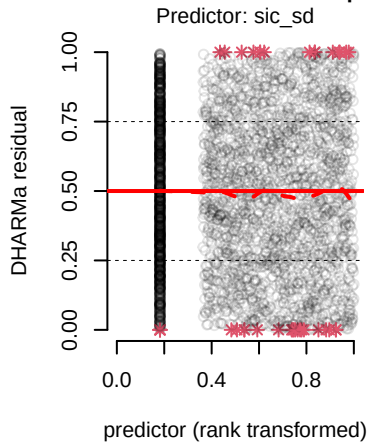
DHARMa residual Residual vs. pred



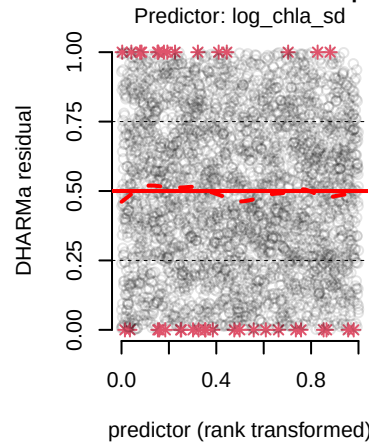
DHARMa residual Residual vs. pred



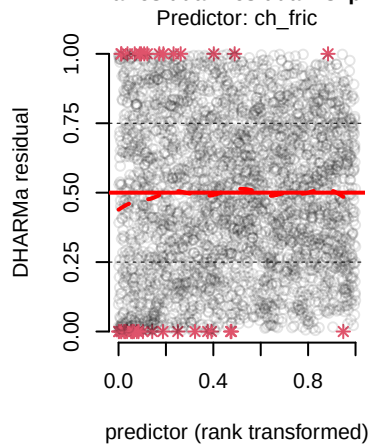
DHARMa residual Residual vs. pred



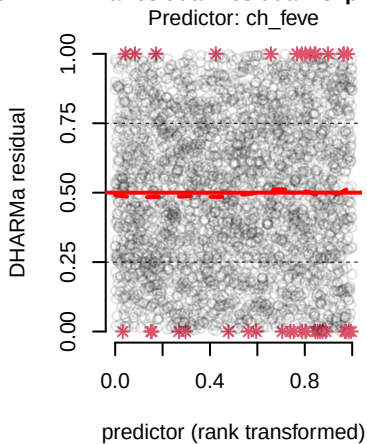
DHARMa residual Residual vs. pred



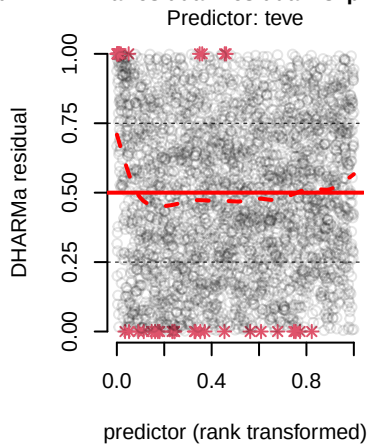
DHARMa residual Residual vs. pred



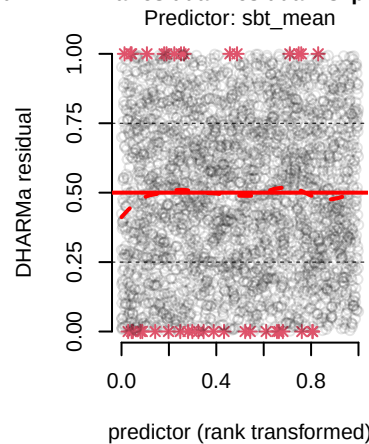
DHARMa residual Residual vs. pred



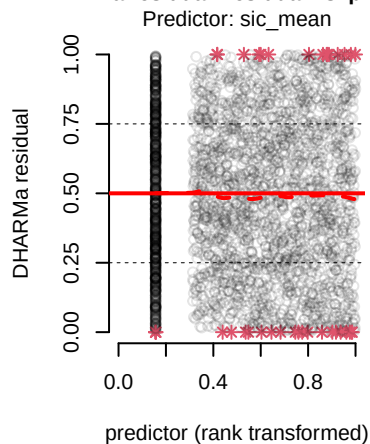
DHARMa residual Residual vs. pred



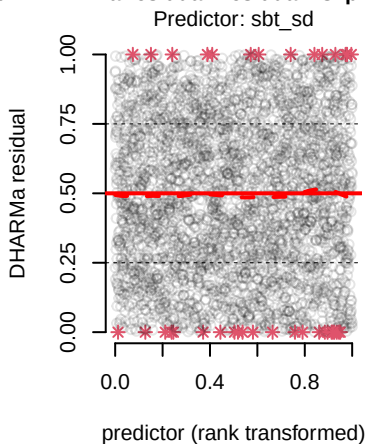
DHARMa residual Residual vs. pred



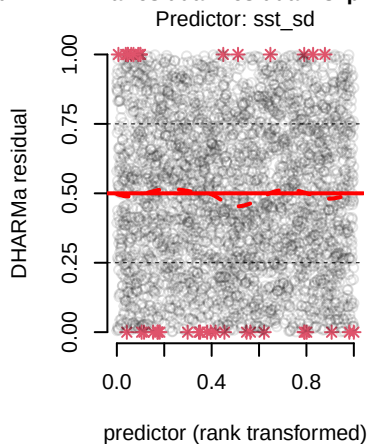
DHARMa residual Residual vs. pred



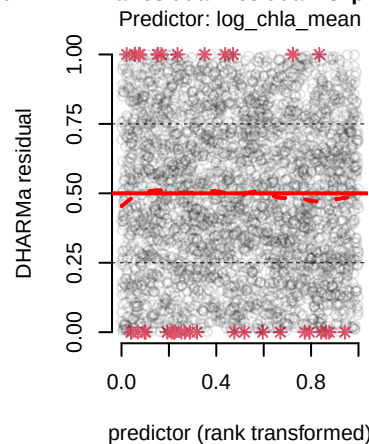
DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred

